Quentin D. Read

Stats and data science to sustainably feed the world

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Professional appointments

2021– **USDA Agricultural Research Service**, Raleigh, NC
Applied consulting statistician (GS-13 as of 2023)
Southeast Area, 2021-2026
Plant Science Research Office, 2026-present

2018-2021 **National Socio-Environmental Synthesis Center**, Annapolis, MD (University of Maryland)
2019–2021 Data scientist
2018-2020 Postdoctoral fellowship: "Food waste and the environment"
Mentor: Dr. Mary Muth (RTI International, Research Triangle Park, NC)

2016-2018 **Michigan State University**, East Lansing, MI
Postdoctoral researcher, Department of Forestry; Ecology, Evolutionary Biology, & Behavior Program
"Intraspecific trait variation and community structure at a continental scale"
Advisors: Dr. Phoebe Zarnetske (MSU) and Dr. Sydne Record (Bryn Mawr)
Visiting scholar, University of Notre Dame, laboratory of Jason McLachlan

Current roles and responsibilities

- Support USDA researchers across the country by designing experiments, processing and visualizing data, and doing statistical analyses
- Develop and teach statistics and data science lessons in virtual and in-person format: see SEAStats training page for lesson text and slides
- Promote modern statistical approaches and best practices for open and reproducible science at USDA
- Co-write manuscripts, presentations, and proposals; review long-term research plans of plant science research units

Education

2011-2016 **University of Tennessee**, Knoxville, TN
Ph.D., Ecology & Evolutionary Biology
Dissertation: "Individual variation in plant traits drives species interactions, ecosystem functioning, and responses to global change"
Advisor: Dr. Nathan Sanders

2005-2009 **University of North Carolina**, Chapel Hill, NC
B.S. with highest distinction, Environmental Science

Skills and languages

- Certified Carpentries data science instructor
- Data processing, visualizing, and analysis in R, including tidyverse and data.table
- Developing R packages
- Bayesian modeling with Stan and brms
- Geospatial data processing and modeling with GDAL and R
- Statistical analysis with SAS 9

- Working knowledge of Python and Julia
- Bash scripting
- Creating static webpages with Markdown and Jekyll
- High-performance parallel computing using Linux server
- Using git/GitHub for version control and remote collaborations
- Fluent in spoken and written German; communicate effectively in spoken and written Spanish

Grants

2019	"Connecting local, regional, and continental scale drivers to biodiversity across NEON through the lens of intraspecific trait variation and disturbance." NSF Macrosystems Biology, NEON-Enabled Science (senior personnel)	$536,800

Publications

OA = open access

2026

Chandrashekhar, R., F. Olawale, B. K. Mulakala, **Q. D. Read**, B. Sutton, S. Lewis, A. Cerney, R. Fox, T. LeRoith, K. K. Gill, B. Nanduri, M. L. Ruebel, E. Lucas, and L. Yeruva. 2026. Pinto bean supplementation improves maternal metabolic outcomes in a high-fat diet-induced obesity model of pregnancy. *The Journal of Nutrition*, 101899. DOI: 10.1016/j.tjnut.2026.101899. (OA)

Bucior, E., **Q. D. Read**, R. B. Sorensen, M. C. Lamb, and C. L. Butts. 2026. Peanut yield and quality not improved by methods to constrain indeterminacy. *Agronomy Journal* 118(6), e70625. DOI: 10.1002/agj2.70625. (OA)

Wang, X., G. T. Billings, A. M. Hulse-Kemp, **Q. D. Read**, Y. Jia, M. H. Jia, Y. Huang, Y. Wamishe, and B. Valent. 2026. Avirulence genes of *Magnaporthe oryzae* are a pathogen defense arsenal shaped by cognate rice resistance genes in the field. *New Phytologist* 252(5), 2043-2055. DOI: 10.1111/nph.71586. (OA)

Paye, W. S., E. Billman, **Q. D. Read**, T. F. Ducey, P. Shumaker, and D. W. Watts. 2026. Soil organic carbon and nitrogen responses to cover crops in Norfolk soil under long-term contrasting tillage systems. *Soil Science Society of America Journal* 90(6), e70380. DOI: 10.1002/saj2.70380. (OA)

Gurung, M., R. Chandrashekhar, J. L. Thomson, A. S. Landry, **Q. D. Read**, A. E. Martinez, A. Andres, and L. Yeruva. 2026. Associations between fermented dairy food intake during pregnancy and gestational weight gain, macrosomia, and gut microbiota. *Clinical Nutrition ESPEN* 76, 105194. DOI: 10.1016/j.clnesp.2026.105194. (OA)

Mulakala, B. K., M. L. Salinas, L. A. Davidson, S. Romero, **Q. D. Read**, R. Fox, T. LeRoith, J. J. Cai, R. S. Chapkin, S. M. Donovan, and L. Yeruva. 2026. *Bifidobacteria infantis* and human milk oligosaccharides have independent and synergistic effects on immune response and amino acid metabolism in germ-free mouse models. *mSystems* 11:00392-26. DOI: 10.1128/msystems.00392-26. (OA)

Pokoo-Aikins, A., T. Chung, A. Oladeinde, **Q. D. Read**, A. Adeli, J. Kwakye, O. Ariyo, A. L. Fuller, A. F. A. Ghareeb, M. C. Milfort, J. P. Lawrence, T. Hamaoka, M. Rothrock, S. E. Aggrey, and Z. Abdo. 2026. Early litter microbiota changes are associated with footpad dermatitis severity in broilers challenged with *Salmonella* Infantis. *Avian Diseases*. DOI: 10.1637/aviandiseases-D-26-00025. (OA)

Habteweld, A., **Q. D. Read**, L. A. Schumacher, H. L. Tyler, P. Mubvumba, N. Bellaloui, H. M. Kelly, and A. Mengistu. 2026. Microbes and nematodes associated with rhizoplane and rhizosphere soil of resistant and susceptible soybean genotypes grown in a *Macrophomina phaseolina*-infested field. *PhytoFrontiers*. DOI: 10.1094/PHYTOFR-05-26-0059-R.

Mengistu, A., **Q. D. Read**, P. Mubvumba, H. M. Kelly, J. Deffenbaugh, A. Habteweld, N. Bellaloui, and L. A. Schumacher. 2026. Charcoal rot severity and soybean yield responses to seeding rate and row spacing. *Plant Disease* 110(8), 3303-3310. DOI: 10.1094/PDIS-09-25-1898-RE.

Valles, S. M., R. L. White, and **Q. D. Read**. 2026. Male and female reproductives are potential sources of RNA virus dissemination in *Solenopsis invicta*. *Journal of Invertebrate Pathology* 219, 108723. DOI: 10.1016/j.jip.2026.108723.

Li, S., J. R. Smith, A. M. Gillen, and **Q. D. Read**. 2026. Multi-year evaluation and release of improved soybean lines for resistance to *Phomopsis* seed decay. *PLoS ONE* 21(8), e0354166. DOI: 10.1371/journal.pone.0354166. (OA)

Harris-Shultz, K. R., V. L. Corby-Harris, G. DeGrandi-Hoffman, L. A. Snyder, E. E. Watkins de Jong, J. S. O'Hearn, **Q. D. Read**, and C. S. Clem. Pollen nutrient content, insect visitation, and morphological characteristics of warm-season turfgrasses. *Journal of Entomological Science* 61(3), 717-730. DOI: 10.18474/JES25-68.

Woolfolk, S. W., D. P. Jeffers, **Q. D. Read,** L. K. Hawkins, and E. M. Babiker. 2026. Assessing varied maize germplasm lines for resistance to fall armyworm and corn earworm (Lepidoptera: Noctuidae) with agronomic quality consideration. *Florida Entomologist* 109(1), 20250041. DOI: 10.1515/flaent-2025-0041.

Lee, H.-S., L. E. Ibarra, B. Zuo, J. Zhao, R. Fox, **Q. D. Read**, M. Gurung, L. Yeruva, C. Chen, and S. P. Trudo. 2026. Protective effect of apiaceous vegetables against Total Western Diet- and dextran sulfate sodium-induced colitis in mice. *The Journal of Nutrition* 156(8), 101652. DOI: 10.1016/j.tjnut.2026.101652.

Copes, W. E., K. R. Leep, S.-E. Lu, **Q. D. Read**, L. Robinson, E. Willis, S. P. Kumpatla, and B. J. Smith. 2026. A survey of the distribution of *Pseudomonas amygdali* pv. *loropetali* present on and in stems of *Loropetalum chinense* shrubs in ornamental nurseries. *Plant Health Progress*. DOI: 10.1094/PHP-01-26-0013-S. *Editor's Pick*. (OA)

Sasidharan, S. K., K. E. C. Elliott, J. D. Evans, **Q. D. Read**, and E. D. Peebles. 2026. Effects of sampling location and frequency on the detection of *Mycoplasma gallisepticum* populations in the respiratory tract of commercial layer pullets. *Poultry Science* 105(9), 107277. DOI: 10.1016/j.psj.2026.107277.

Ruebel, M., Y. Zhong, P. Z. Schall, P. Jambal, D. N. Yazza, K. Zemski-Berry, S. V. Chintapalli, **Q. D. Read**, K. M. Thakali, and K. Shankar. 2026. MyD88 signaling in trophoblasts contributes to maternal high fat diet-associated alterations in placental gene expression and lipid profiles. *Placenta* 182, 79-90. DOI: 10.1016/j.placenta.2026.06.001.

Nour, I., J. R. Blakey, S. Alvarez-Narvaez, A. Kulkarni, **Q. D. Read**, and S. K. Mohanty. 2026. Novel multigene molecular characterization of avian reovirus strains and associated embryonic pathogenicity. *Journal of Virology*, e01982-25. DOI: 10.1128/jvi.01982-25.

Garcia, J. S., D. R. Jones, R. K. Gast, **Q. D. Read**, V. Hardman, and R. Meinersmann. 2026. Applied Research Note: Impact of egg handling and storage temperature on the growth of spoilage organisms in shell eggs stored for 27 weeks. *Journal of Applied Poultry Research*, 100730. DOI: 10.1016/j.japr.2026.100730. (OA)

Pokoo-Aikins, A., T. R. Mitchell, **Q. D. Read**, R. Shanmugasundaram, and A. E. Glenn. 2026. Mycotoxin occurrence in poultry feed and feed ingredients. *Journal of Applied Poultry Research*, 100729. DOI: 10.1016/j.japr.2026.100729. (OA)

Poudel, I., T. C. Vu, R. Wysocky, R. F. Malheiros, **Q. D. Read**, K. E. Anderson, and O. T. Toomer. 2026. Incorporating high-oleic peanuts in layer diets: impact

on production, egg quality & nutritional profile and calculated egg income and feed cost. *Journal of Applied Poultry Research* 35(3), 100719. DOI: 10.1016/j.japr.2026.100719. (OA)

McIntyre, J. S., H. J. Cook, Q. D. Read, R. B. Sorensen, E. R. Bucior, and M. C. Lamb. 2026. Diminished shell strength from rainfall when peanuts are in windrows at harvest. *Peanut Science* 53(1), 31-43. DOI: 10.3146/0095-3679-53-PS1665. (OA)

Richards, A. K., E. A. McMillan, **Q. D. Read**, N. W. Shariat, R. J. Buhr, and C. E. Harris. 2026. Evaluation of MicroTally mitts for detecting foodborne pathogens during broiler processing. *Journal of Food Protection* 89(5), 100740. DOI: 10.1016/j.jfp.2026.100740. (OA)

Kannan, N. and **Q. D. Read**. 2026. Selection of the least drifting spray nozzles for pesticide application to protect insect pollinators. *Journal of the ASABE* 69(1): 77-91. DOI: 10.13031/ja.16415.

R. L. White, **Q. D. Read**, and S. M. Valles. 2026. Evaluation of organosilicone surfactants as synergists of Solenopsis invicta virus 3 in *Solenopsis invicta*. *Journal of Invertebrate Pathology* 216, 108573. DOI: 10.1016/j.jip.2026.108573.

Brinkhoff, R. E., N. J. Sanders, J. A. Henning, G. Newman, **Q. D. Read**, M. K. Sundqvist, M. J. Hovenden, C. M. Prager, K. E. Rewcastle, L. Souza, O. K. Vought, and A. T. Classen. 2026. The impact of warming on peak-season ecosystem carbon uptake is influenced by dominant species in warmer sites. *Journal of Ecology* 114(2), e70247. DOI: 10.1111/1365-2745.70247.

Jia, J., K. Leep, K. Phillips, E. Willis, L. Robinson, W. Phillips, **Q. D. Read**, W. E. Copes, and S. Lu. 2025. Development of a quantitative PCR method to detect the bacterial gall pathogen *Pseudomonas amygdali* pv. *loropetali* from loropetalum plant materials. *Plant Disease*, PDIS-11-23-2533-SR. DOI: 10.1094/PDIS-11-23-2533-SR.

Billings, G. T., D. Restrepo-Montoya, L. A. Dentinger, J. W. Zirkel, B. T. Campbell, D. C. Jones, **Q. D. Read**, P. W. Chee, B. E. Scheffler, C. H. Haigler, A. M. Hulse-Kemp, and J. A. Scheffler. 2026. The cotton historical lines project – Part I: History and field evaluation. *Crop Science* 66(1), e70173. DOI: 10.1002/csc2.70173. (OA)

2025 Garcia, J. S., R. K. Gast, **Q. D. Read**, and D. R. Jones. 2025. The impact of egg handling and storage temperature on shell eggs stored for 27 weeks: Egg quality. *Poultry Science* 105(2), 106264. DOI: 10.1016/j.psj.2025.106264.

Harrell, T., S. Alvarez-Narvaez, **Q. D. Read**, and S. J. Conrad. Novel strand-specific qPCR assay elucidates differences in viral replication kinetics between different strains of avian reovirus. *Microbiology Spectrum* 13, e03140-24. DOI: 10.1128/spectrum.03140-24. (OA)

McKenzie-Reynolds, P., I. A. Owolabi, A. Biswas, **Q. D. Read**, G. R. Burke, A. Levi, and A. M. Simmons. 2025. Wild sources for host plant resistance to *Bemisia tabaci* in watermelon: insights from behavioral and chemical analyses. *Crop Protection* 199, 107411. DOI: 10.1016/j.cropro.2025.107411.

Harris-Shultz, K., J. Wallace, S. Sapkota, B. Schwartz, **Q. D. Read**, J. Peterson, A. L. Santana, D. Smolensky, and A. Coffin. 2025. Identification of candidate genes for stigma color using a genome-wide association study in centipedegrass. *Crop Science* 65(6), e70185. DOI: 10.1002/csc2.70185. (OA)

Cottrell, T. E., **Q. D. Read**, and C. Starns. 2025. Electron beam irradiation for management of in-shell pecan weevil larvae (Coleoptera: Curculionidae). *Journal of Economic Entomology*, toaf300. DOI: 10.1093/jee/toaf300.

Toomer, O. T., T. C. Vu, R. Wysocky, V. Moraes, R. Malheiros, **Q. D. Read**, and K. E. Anderson. 2025. Effects of an unblanched peanut and/or peanut skin diet on

egg quality, egg lipid chemistry, and performance of hens housed in a cage-free environment. *Animals* 15(18), 2673. DOI: 10.3390/ani15182673. (OA)

Oakley, B. A., T. R. Mitchell, **Q. D. Read**, G. T. Hibbs, T. T. Baldwin, L. J. Pierce, S. E. Gold, and A. E. Glenn. 2025. A nitric oxide reductase is a key enzyme target for eliminating fungal emissions of nitrous oxide. *Fungal Genetics and Biology*, 104038. DOI: 10.1016/j.fgb.2025.104038.

McMillan, E., **Q. D. Read**, T. Mitchell, C. Harris, A. E. Glenn, A. Pokoo-Aikins, R. Meinersmann, and J. Frye. 2025. *In vitro* synergistic effects of peracetic acid and biocides approved for use in poultry meat processing against four serotypes of non-typhoidal *Salmonella enterica*. *Microbiology Spectrum*: e00652-25. DOI: 10.1128/spectrum.00652-25.

Chen, C. and **Q. D. Read**. 2025. Variability of ripening peaches from different orchards, seasons, and harvests in the southeastern USA. *Technology in Horticulture* 5, e032. DOI: 10.48130/tihort-0025-0027. (OA)

Portilla, M., N. S. Little, B. H. Elkins, Y. Du, Y. C. Zhu, J. P. Glover, and **Q. D. Read**. 2025. Evaluation of *Beauveria bassiana* strain NI8 and conventional insecticides to control *Lygus lineolaris* in cotton. *Southwestern Entomologist* 50(2): 611-631. DOI: 10.3958/059.050.0225.

Eichenwald, A., J. M. Grady, J. Knott, **Q. D. Read**, J. M. Rodriguez, and S. Record. 2025. The impact of disturbance on tree size distributions in the United States. *Global Ecology & Biogeography* 34(8): e70102. DOI: 10.1111/geb.70102.

Valles, S. M., M. S. Ascunce, **Q. D. Read**, R. L. White, and R. K. Vander Meer. 2025. Virome of *Solenopsis invicta* (Hymenoptera: Formicidae): Intercontinental differences in the native and introduced ranges. *Journal of Invertebrate Pathology* 213: 108401. DOI: 10.1016/j.jip.2025.108401. (OA)

Oppenheimer, P., F. Tini, R. Whetten, I. Laraba, **Q. D. Read**, B. Whitaker, M. Vaughan, G. Beccari, L. Covarelli, and C. Cowger. 2025. Synthetic spike-in metabarcoding for plant pathogen diagnostics results in precise quantification of copy number within the genus *Fusarium*. *International Society of Microbial Ecology Communications*, ycaf124. DOI: 10.1093/ismeco/ycaf124. (OA)

Pawlowski, E. D., L. Witthaus, J. M. Taylor, M. T. Moore, E. Stevens, S. McNamara, and **Q. D. Read**. 2025. The role of vegetation, drying, and rewetting on water and sediment phosphorus concentrations in an experimental ditch channel experiment. *Agrosystems, Geosciences & Environment* 8(3): e70168. DOI: 10.1002/agg2.70168. (OA)

Penn, H. J., **Q. D. Read**, R. T. Richard, and D. J. Dufrene. 2025. Prior stem borer damage does not impact conspecific damage or perennial crop yield in subsequent years of growth. *Entomologia Experimentalis et Applicata*, e13609. DOI: 10.1111/eea.13609.

Paye, W. S., R. Moral, M. B. Vanotti, A. A. Szögi, and **Q. D. Read**. 2025. Agronomic effectiveness of nitrogen and phosphorus recovered from swine manure. *Agrosystems, Geosciences & Environment* 8(2): e70153. DOI: 10.1002/agg2.70153. (OA)

Copes, W. E., **Q. D. Read**, and B. J. Smith. 2025. Application variables that affect efficacy of disinfestants sprayed on different substrate materials to control *Colletotrichum siamense*. *PhytoFrontiers* 5(3). DOI: 10.1094/PHYTOFR-12-24-0131-R. (OA)

Ruebel, M., S. P. Gilley, P. Jambal, J. Dado-Fox, D. N. Yazza, N. Nakra, C. Uhlson, L. Sian, S. Kode, **Q. D. Read**, L. Yeruva, J. L. E. Westcott, P. S. MacLean, N. F. Krebs, and K. Shankar. 2025. Maternal undernutrition exacerbates effects of ambient heat during pregnancy in mice. *The Journal of Nutrition* 155(9): 2996-3007. DOI: 10.1016/j.tjnut.2025.05.021.

Chatterjee, A., J. M. Taylor, **Q. D. Read**, M. T. Moore, M. A. Locke, and J. D. Hoeksema. 2025. Water quality and soil nutrient availability trade-offs associated with timing and duration of managed flooding for migratory waterbird habitat. *Soil Science Society of America Journal* 89(3): e70077. DOI: 10.1002/saj2.70077.

Newman, C., R. Austin, R. Andres, **Q. D. Read**, N. Garrity, K. Fritz, A. Oakley, A. Hulse-Kemp, and J. Dunne. 2025. Evaluating UAV captured RGB and multispectral imagery as a proxy for visual rating of leaf spot in cultivated peanut. *The Plant Phenome Journal* 8(1): e70019. DOI: 10.1002/ppj2.70019. (OA)

Rodrigues, N. B., T. R. Barbosa, H. S. Koenow Pinheiro, M. Mancini, **Q. D. Read**, J. Blackstock, H. E. Winzeler, D. Miller, P. R. Owens, and Z. Libohova. 2025. Influences of sampling design and model selection on predictions of chemical compounds in Petroferric formations in the Brazilian Amazon. *Remote Sensing* 17(9): 1644. DOI: 10.3390/rs17091644. (OA)

Kannan, N., B. Fritz, Q. D. Read, and R. Srinivasan. 2025. Testing the drift reduction potential of some adjuvants from renewable sources for aerial pesticide applications. Book chapter *in* Pesticide Formulation and Delivery Systems: 43rd Volume, Creating Certainty in an Uncertain World, ed. Danny Brown (West Conshohocken, PA: ASTM International, 2025), 134–147. DOI: 10.1520/STP165220240015.

Woolfolk, S., G. A. Matthews, and **Q. D. Read**. 2025. Comparison of infestation rates of fall armyworm (Lepidoptera: Noctuidae) neonates for maize resistance screening. *Journal of Insect Science* 25(2): 9. DOI: 10.1093/jisesa/ieaf025. (OA)

Mengistu, A., **Q. D. Read**, C. R. Little, H. M. Kelly, and N. Bellaloui. 2025. Severity of charcoal rot disease in soybean genotypes inoculated with *Macrophomina phaseolina* isolates differs among growth environments. *Plant Disease* 109(10). DOI: 10.1094/PDIS-10-24-2230-RE.

Smith, B. J., E. T. Stafne, and **Q. D. Read**. 2025. Establishment of southern highbush blueberry cultivars and suppression of *Phytophthora* root rot using cover crop and soil amendment treatments. *PhytoFrontiers* 5(3). DOI: 10.1094/PHYTOFR-09-24-0096-R. (OA)

Caren, J., Y.-C. Zhu, **Q. D. Read**, and Y. Du. 2025. Risk assessment of essential oils on honey bees (*Apis mellifera* L.). *Insects* 16(3):303. DOI: 10.3390/insects16030303. (OA)

Adeli, A., J. P. Brooks, D. M. Miles, **Q. D. Read**, Y. Huang, G. Feng, and J. N. Jenkins. 2025. Integrated effects of tillage, fertilizer sources and a rye cover crop on dryland cotton production. *Agronomy Journal* 117(2):e70019. DOI: 10.1002/agj2.70019.

Lofton, L. W., **Q. D. Read**, H. L. Hamilton, A. E. Glenn, J. A. Hawkins, T. R. Mitchell, and S. E. Gold. 2025. Pyrrocidines A and B demonstrate synergistic inhibition of *Fusarium verticillioides* growth. *Frontiers in Microbiology* 15. DOI: 10.3389/fmicb.2024.1480920. (OA)

Yeh, H.-Y. and **Q. D. Read**. 2025. Immune responses of chickens against recombinant *Salmonella enterica* serotype Heidelberg FimA and FimW fimbriae and FliD and FlgK flagellar proteins. *Veterinary Immunology and Immunopathology* 280:110870. DOI: 10.1016/j.vetimm.2024.110870.

2024 Oi, D. H., R. A. Atchison, and **Q. D. Read**. 2024. Effect of fast and slow-acting bait toxicants on tawny crazy ant (Hymenoptera: Formicidae) foraging and nesting in the field. *Journal of Economic Entomology* toae284. DOI: 10.1093/jee/toae284.

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Frazier, S., S. M. Brown, **Q. D. Read**, A. L. Jacobson, K. Conner, C. Escalante, and K. S. Balkcom. 2024. Cotton stalk management and a cover crop produce minimal effects on cotton leafroll dwarf virus. *Agronomy Journal*. DOI: 10.1002/agj2.70002. (OA)

Johnson, R. M., K. A. Richard, and **Q. D. Read**. 2024. Effects of potassium fertilizer on sugarcane yields and plant and soil potassium levels in Louisiana. *Agronomy* 14(12): 2761. DOI: 10.3390/agronomy14122761. (OA)

Kim, S.-B., K.-T. Kim, S. In, N. Jaiswal, G.-W. Lee, S. Jung, A. Rogers, L. F. Gómez-Trejo, S. Gautam, M. Helm, H.-K. Ahn, H.-Y. Lee, **Q. D. Read**, J. Woo, K. L. Holan, S. A. Whitham, J. D. G. Jones, D. Choi, R. Dean, E. Park, and P. Balint-Kurti. 2024. Use of the *Puccinia sorghi* haustorial transcriptome to identify and characterize AvrRp1-D recognized by the maize Rp1-D resistance protein. *PLoS Pathogens* 20(11): e1012662. DOI: 10.1371/journal.ppat.1012662. (OA)

Assumpcao, A. L. F. V., K. Arsi, K. S. Alharbi, A. Asnayanti, A. D. Trieu, **Q. D. Read**, R. Perera, A. Shwani, A. Hasan, S. Pillai, A. M. Donoghue, D. Rhoads, P. R. R. Jesudhasan, and A. A. K. Alrubaye. 2024. Electron-beam-killed *Staphylococcus* vaccine reduced lameness in broiler chickens. *Vaccines* 12(11): 1203. DOI: 10.3390/vaccines12111203. (OA)

McMillan, E., E. S. Adams, T. Mitchell, J. Hawkins, **Q. D. Read**, A. Pokoo-Aikins, M. Berrang, C. E. Harris, M. D. Hughes, A. E. Glenn, and R. J. Meinersmann. 2024. Susceptibility of pESI positive Salmonella to treatment with biocide chemicals approved for use in poultry processing as compared to Salmonella without the pESI plasmid. *Letters in Applied Microbiology* 77(7): ovae067. DOI: 10.1093/lambio/ovae067.

H. Abou-Shaara, S. Mehrparvar, **Q. D. Read**, J. Chen, and E. Amiri. 2024. Impact of commercial plastic queen cell cups on rearing success and development of honey bee queens. *Journal of Apicultural Research* 64(4): 1074–1084. DOI: 10.1080/00218839.2024.2418682.

Pokoo-Aikins, A., C. M. McDonough, T. R. Mitchell, J. A. Hawkins, L. F. Adams, **Q. D. Read**, X. Li, R. Shanmugasundaram, E. Rodewald, P. Acharya, A. E. Glenn, and S. E. Gold. 2024. Mycotoxin contamination and the nutritional content of corn targeted for animal feed. *Poultry Science* 103(12): 104303. DOI: 10.1016/j.psj.2024.104303.

Libohova, Z., M. Mancini, H. E. Winzeler, **Q. D. Read**, N. Sun, D. Beaudette, C. Williams, J. Blackstock, S. H. G. Silva, N. Curi, K. Adhikari, A. Ashworth, J. Minia, and P. R. Owens. 2024. Interpreting the spatial distribution of soil properties with a physically-based distributed hydrological model. *Geoderma Regional*: e00863. DOI: 10.1016/j.geodrs.2024.e00863. (OA)

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Romano, N., P. Wills, J. Paredes, L. Chin, C. Perricone, **Q. D. Read**, W. Sealey, and J. Perez. Influence of dietary guar gum on pompano (*Trachinotus carolinus*) growth, feeding efficiency, liver/intestinal histopathology, and solids removal efficiency using extruded pellets. *Aquaculture Reports*, submitted.

Ruebel, M. L, B. K. Mulakala, E. C. Diaz, C. M. Burnham, **Q. D. Read**, E. Borsheim, and L. Yeruva. Metagenomic and metabolomic profiling reveals metabolic markers link to excess adiposity - The Arkansas Active Kids study. *The American Journal of Clinical Nutrition*, submitted.

Woolfolk, S. W., W. Xu, **Q. D. Read**, and E. M. Babiker. New resistance sources against fall armyworm and corn earworm (Lepidoptera: Noctuidae) identified via multi-year evaluation of maize and maize-*Tripsacum*/teosinte hybrids. *Journal of Insect Science*, in revision.

Balkcom, K. S., **Q. D. Read**, S. A. Prior, A. L. Jacobson, G. B. Runion, K. Conner, and A. Sanz-Saez. Enriched carbon dioxide did not eliminate negative effects of cotton leafroll dwarf virus. *PhytoFrontiers*, submitted.

Gurung, M., B. Mulakala, J. S. Kumar, R. Gibson, C. Burnham, K. Gill, B. Nanduri, S. Trudo, M. Ruebel, **Q. D. Read**, L. Mora, K. Rodriguez, K. Stephens, and L. Yeruva. Maternal diet-antibiotic interactions reshape placental and fetal brain transcriptomes with sex-specific consequences in offspring. *mSystems*, submitted.

Media articles

Read, Q. D.; J. Pitt, editor. 2022. Extinction on our plates. 360info Special Report from Covering Climate Now's joint coverage week on Food & Water. DOI: 10.54377/42f4-f24c.

Teaching and curriculum development

2026	Designed and taught two-day workshop for USDA researchers: "Introduction to multiomics integration"
	Designed and taught one-day workshop for USDA researchers: "Dealing with missing data"
2025	Designed and taught one-day workshops for USDA researchers: "Machine learning demystified"; "Power analysis tutorial"
2024	Designed and taught workshops for USDA researchers: "Bayesian generalized linear mixed models"; "R for SAS Users"; "A smorgasbord of options for multiomics data analysis"
	Designed and presented lectures for USDA researchers on interpreting statistical interactions, means comparison tests, dealing with common errors in statistical model fitting, and analyzing categorical phenotype data

2023	Designed and taught lessons for USDA researchers: “Bayesian mixed models with brms,” “Data visualization basics with R and ggplot2”
2022	Designed and taught two-day workshop for USDA researchers: “A practical toolkit for mixed models in R”
2021	Co-designed curriculum for day-long geospatial data analysis workshop for postdocs and grad students; taught introductory lesson and lesson on geospatial statistics with vector data
2020	Co-designed and co-taught day-long whirlwind data science class for SESYNC postdoctoral researchers
2020-2021	Co-taught SESYNC 2020 and 2021 Computational Summer Institute, a week-long online data science course
	Designed and led workshop on best practices for collaboration with GitHub
2018	Designed graduate teaching module: “Using NEON data to test macroecological hypotheses”
2017	Co-instructor of record, Metacommunity Ecology, MSU
2016	Graduate teaching assistant, Models in Biology, UT-Knoxville
2015-2016	Graduate teaching assistant, Conservation Biology, UT-Knoxville
2015	Designed and led workshop on graphing with R and ggplot2
2014	Designed and led workshops on introductory statistics with R
2013-2014	Member of panel developing and reforming curriculum of undergraduate introductory biology courses, UT-Knoxville
2012-2014	Graduate teaching assistant, General Ecology, UT-Knoxville
2011-2012	Graduate teaching assistant, Introduction to Biodiversity, UT-Knoxville

Mentoring

2024-2026	Member of Nirwan Tandukar’s graduate committee, N.C. State University Department of Molecular and Structural Biochemistry
2021	Mentored student team in University of Maryland Data Challenge; team won grand prize in a 75-team competition
2020	Remotely mentored two Bryn Mawr College undergraduates on an R package development project, funded by NSF Macrosystems Biology grant
2017-2018	Mentored 4 undergraduates through Summer Research Opportunities Program and High Performance Computing Center, MSU
2012-2015	Mentored 8 undergraduate field and lab assistants, Rocky Mountain Biological Laboratory
2013-2015	Mentored 3 undergraduate lab assistants, UT-Knoxville

Fellowships and awards

2024	USDA-ARS Southeast Area Annual Employee Recognition Program: award for Customer Service and Technical Expertise	
2018	NSF SESYNC postdoctoral immersion fellowship	$160,000
2015	University of Tennessee Science Alliance graduate award	$3000
	NSF travel grant to deliver biology education seminar at Notre Dame	$500

2014	Outstanding Outreach and Community Service award, UT-Knoxville Department of Ecology and Evolutionary Biology	
2013	Dr. Jean H. Langenheim Endowed Graduate Fellowship in the Ecology and Evolution of Plants, Rocky Mountain Biological Laboratory	$6000
	UTK Graduate Student Senate travel award	$500
	Honorable Mention, National Science Foundation Graduate Fellowship	
2012	Dr. Lee R. G. Snyder Memorial Fellowship, Rocky Mountain Biological Laboratory	$700

Invited talks

V = virtual

2026 "Designing animal experiments: old and new paradigms." USDA ARS 2026 IACUC workshop, Auburn, AL (V)

"Power analysis basics." USDA ARS Delta Water Management Research Unit, Jonesboro, AR

2025 "Multinomial logistic GLMMs: a patchy landscape of software implementations." AgStats (Conference on Applied Statistics in Agriculture & Natural Resources), as part of session I co-organized on GLMM best practices, Gainesville, FL (V)

"Experimental design matters!" USDA ARS coffee hour seminar series, New Orleans, LA (V)

2024 "Doing more with less: using prior knowledge and creative experimental design to get more results from fewer animals." USDA ARS 2024 IACUC workshop, Stuttgart, AR (V)

"Analyzing ordered categorical phenotypes: challenges and pitfalls." USDA ARS SIBS Phenomics working group meeting, Raleigh, NC

"Troubleshooting common errors and warnings in (G)L(M)Ms." USDA ARS Honey Bee Lab, Baton Rouge, LA

"Statistical interactions: what are they and what do they mean, anyway?" USDA ARS Southern Regional Research Center, New Orleans, LA

"Everything you ever wanted to know about means comparisons but were afraid to ask." USDA ARS Dale Bumpers National Rice Research Center, Stuttgart, AR

"A smorgasbord of options for multiomics data analysis." USDA ARS and Arkansas Children's Nutrition Center, Little Rock, AR

2023 "Structural equation modeling in food science." USDA ARS Food Science & Market Quality & Handling Research Unit lab meeting, Raleigh, NC

"Power analysis: bureaucratic busywork or critical step in the scientific method?" USDA ARS 2023 IACUC workshop, Athens, GA (V)

2022 "Reducing food waste from farm to fork to benefit biodiversity." North Carolina State University, Plant & Microbial Biology department seminar, Raleigh, NC

2020 "Reducing food waste to benefit environment and society: how, why, and in what context?" Duke University Program in Ecology seminar series, Durham, NC (V)

2018 "Food waste impacts on biodiversity." Commission on Environmental Cooperation Food Loss and Waste Measurement Experts Group Meeting, Arlington, VA

2017 "Challenges in the functional trait approach to community ecology (and ways to overcome them)." German Centre for Integrative Biodiversity Research (iDiv), Leipzig, Germany

"Intraspecific variation and community assembly." National Ecological Observatory Network, Boulder, CO (talk co-delivered with John Grady)

2016 "Individual variation in organismal traits: predicting patterns in space and time from local to global scales." Michigan State University Department of Forestry, Hanover Forest Science Seminar Series, East Lansing, MI

2015 "C3UBE undergraduate biology curriculum reform." University of Notre Dame, Biology Education Seminar, Notre Dame, IN

2014 "Roots, leaves, and soils facing global change." Rocky Mountain Biological Laboratory Seminar, Gothic, CO

"Plant traits & interactions altered by warming at different elevations." Oak Ridge National Laboratory, Environmental Sciences Division, Oak Ridge, TN

Conference presentations

2024 **Read, Q. D.** and F. Breidt. "Exploring the options for quantifying uncertainty in derivatives of splines." NCCC-170 meeting, Mayagüez, PR

2023 **Read, Q. D.** and H. J. Penn. "'Boring' for insights in a 28-year sugarcane insect herbivory dataset." AgStats, West Lafayette, IN

2019 **Read, Q. D.**, J. A. Gephart, A. D. Cuéllar, … and M. K. Muth. "Which supply chain stages should we target to reduce food loss and waste and benefit the environment?" Ecological Society of America-United States Society for Ecological Economics, Louisville, KY

2018 **Read, Q. D.**, S. Record, K. M. Dahlin, P. L. Zarnetske, and others. "Measuring geodiversity to explain biodiversity: what is the effect of spatial grain and spatial consciousness?" US-International Association of Landscape Ecology, Chicago, IL

2017 **Read, Q. D.**, J. M. Grady, P. L. Zarnetske, S. Record, and others. "Intraspecific variation reflects drivers of rodent community assembly across the National Ecological Observatory Network." Ecological Society of America, Portland, OR

2015 **Read, Q. D.**, N. J. Sanders, and A. T. Classen. "A globally replicated experiment shows that long-term environmental filters constrain plant response to increased temperature and loss of foundation species." Ecological Society of America, Baltimore, MD

Guest lectures

2026 Ethical and responsible statistical practices, Environmental Data Analysis (undergraduate course), William Peace University, Raleigh, NC

2023 Food waste, Global Environmental Issues (undergraduate course), William Peace University

Ethical and responsible statistical practices, Bioethics (graduate/undergraduate course), N.C. State University

2022 Food waste, Agroecology (undergraduate course), University of Maryland

2021 Food waste, People, Land and Food (undergraduate course), George Washington University, Washington, DC

2020 Data synthesis in food-energy-water nexus research, Global Stewards (graduate seminar), University of Maryland

2015 Mock trial activity, Conservation Biology, UT-Knoxville

2013 Climate change and communities, General Ecology, UT-Knoxville

2012 Biogeochemistry, General Ecology, UT-Knoxville

Software

2022 Weinroth, M. and **Q. D. Read.** 2022. epi2me2r: Process Nanopore EPI2ME Output for Use in R. R package version 0.1.0. https://mweinroth.github.io/epi2me2r/.

2021 **Read, Q. D.**, A. Yue, I. E. Fluck, B. Baiser, J. M. Grady, P. L. Zarnetske, and S. Record. 2021. Ostats: O-stats, or pairwise community-level niche overlap statistics. R package version 0.1.1. https://neon-biodiversity.github.io/Ostats/. DOI:10.5281/zenodo.5706470.

2020 Brunson, J. C. and **Q. D. Read.** 2020. ggalluvial: Layered Grammar for Alluvial Plots. R package version 0.12.2.0001. https://corybrunson.github.io/ggalluvial/.

2019 Marchand, P., I. T. Carroll, M. Smorul, R. E. Blake, and **Q. D. Read.** 2019. rslurm: Submit R Calculations to a 'Slurm' Cluster. R package version 0.6.1. https://sesync-ci.github.io/rslurm. DOI:10.5281/zenodo.5705430.

Online content

2019-2021 Blog posts on the SESYNC Cyberhelp blog (https://sesync-ci.github.io/blog)

- “Making a fifty-state USA map, 2021 edition”
- “Making free maps with R, ggspatial, and Mapbox”
- “Goodbye %>%, hello := (Using R data.table to speed up my data science)”
- “How open reproducible methods benefit the research community: a shiny story”
- “The carbon footprint of R code, and how to reduce it”
- “Resources to help you learn GitHub Pages”
- “Tips for a smooth R(Studio) workflow and reproducible R code”
- "How do I resolve merge conflicts in git/GitHub/GitLab?"
- "Using the rslurm package to run code in parallel"
- “ggplot tricks not to forget about”

2021 Marchand, P., I. T. Carroll, and **Q. D. Read**. “Introduction to Geospatial Data.” Jul. 2021, SESYNC Cyberhelp online lesson. https://sesync-ci.github.io/geospatial-packages-in-R-lesson/

2020 **Read, Q. D.** “Advanced git Techniques.” Sep. 2020, SESYNC Cyberhelp online lesson. https://sesync-ci.github.io/advanced-git-lesson/

Read, Q. D. and I. T. Carroll. “Online Data with R.” Jul. 2020, SESYNC Cyberhelp online lesson. https://sesync-ci.github.io/online-data-with-R-lesson/

Muth, M. K. and **Q. D. Read.** “Effects of COVID-19 meat and poultry plant closures on the environment and food security.” 7 Jul. 2020, RTI Insights blog. https://www.rti.org/insights/covid-19-effect-meat-supply-chain

Blake, R. E., R. Beilinson, N. Motzer, K. L. Hondula, and **Q. D. Read.** “Resources and tips for elevating your team science in an all-virtual world.” Mar. 2020. https://www.sesync.org/resources-and-tips-for-elevating-your-team-science-in-an-all-virtual-world

Professional service

2025– Member, USDA ARS Data Management Working Group; providing policy and best-practices recommendations to the agency about data management

2024– Member, NCCC-170 "Research Advances in Agricultural Statistics" research coordinating committee

2024– Member, USDA SciNet (high-performance computing network) Advisory Committee; chair of Education subcommittee

2020– Peer reviewer of R packages for RopenSci.org

2019–2023 Maintainer of the R package *rslurm* (earthdatascience.org/rslurm)

2019 Review panelist, SESYNC immersion postdoctoral fellowship program

2017 Co-organizer of oral session at Ecological Society of America meeting: "Challenges and opportunities for investigating ecological communities across space and time: insights from coordinated research networks."

2013– Peer reviewer for >40 scholarly journals

Community outreach

2023 Panelist, federal careers discussion panel, NC State University biotechnology undergraduate course

2021 Interviewed on food waste for SESYNC podcast (anchor.fm/sesync-communications)

2018 Organized event and gave presentation on citizen science opportunities, MSU Science Festival

2017 Gave public research talk, MSU Biology On Tap

2014 Organized and coordinated Darwin Day Tennessee

2013 Coordinated advertising for Darwin Day Tennessee

2012 Discussed my research and assisted 7th-grade students with climate change projects

2011-2015 Volunteered at kids' science education events at the University of Tennessee and the Rocky Mountain Biological Laboratory

Professional training

2024 Bayesian Procedures in SAS training course

2021 SAS for R Users training course

2020 ReproHack reproducible science workshop, SESYNC

2019 Google Earth Engine workshop, SESYNC

Ecosystem services valuation workshop, USDA, Washington, DC

2018 Data to Motivate Synthesis workshop, SESYNC

2018-2020 SESYNC postdoctoral immersion series, including multiday workshops on interdisciplinary techniques, team science, socio-environmental hydrology, land system science, methods in social science, ecology, environmental policy, and public health

2014 Structural equation modeling workshop with Dr. Jim Grace, Knoxville, TN

2013 Short course: Boreal Forest Ecology, Swedish University of Agricultural Sciences, Umeå, Sweden

2012 Short course: Fundamentals of Ecosystem Ecology, Cary Institute of Ecosystem

Studies, Millbrook, NY